MIL-A-17682E
11 March 1982
SUPERSEDING
MIL-A-17682D
13 March 1975

MILITARY SPECIFICATION

ADHESIVE, STARCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for one type of adhesive used for mounting paper targets.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

N-F-481	- Flour, Wheat.
UU-B-36	- Bag, Paper (Grocers)
UU-P-121	- Paper, Bond and Writing, White and Colored.
PPP-B-566	- Box, Folding, Paperboard.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-76	- Tape, Packaging Paper (for Carton Sealing).

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 8040

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-10831 - Paper, Target.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E7077 - PH of Aqueous Solutions with the Glass Electrode, Test Method For.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

D903 - Peel or Stripping Strength of Adhesive Bonds.
G 2170 - Determining Resistance of Synthetic Polymeric Material to Fungi, Practice for.

(Application for copies should be addressed to American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval. The first article shall be a preproduction sample or it may be a standard production item from the contractor's current inventory.

3.2 Material. The adhesive shall be made from materials specified herein. The adhesive may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed material shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials orginally selected for use in the adhesive.

3.3 Chemical and physical properties.

3.3.1 Composition. The base material for the adhesive shall be made of pure wheat, in fine powder form, conforming to N-F-481, containing sufficient preservatives to prevent decomposition, weevil and mold growth in package, and to prevent fermentation after mixing with water.

3.3.2 Hydrogen-ion concentration (pH) value. The pH value shall be between 3.6 and 7.0 when tested as specified in 4.5.2.

3.3.3 Odor. The adhesive shall be free from musty and other undesirable odors when tested as specified in 4.5.3.

3.3.4 Toxicity. The adhesive must not present a health hazard to individuals from inhalation, skin contact, or ingestion, when used for its intended purpose or as a result of exposure after its application. If toxic or hazardous substances are used in the manufacturing of this product, they shall be identified by a precautionary label of warning. The manufacturer shall furnish the contracting officer with a certificate stating that the adhesive, when applied in accordance with the manufacturer's instructions, is nontoxic (see 4.5.4).

3.3.5 Brushing properties. The adhesive shall be of brushing consistency throughout the temperature range of 50°F to 100°F (10°C to 38°C) when tested as specified in 4.5.5.

3.3.6 Color and transparency. When mixed, the adhesive shall be a light-colored, translucent liquid. After application to targets, it shall dry to a substantially colorless, translucent finish (see 4.5.11).

3.3.7 Homogeneity. The adhesive shall be produced, utilizing the proper blending of materials, to provide a mixture of uniform structure and composition throughout (see 4.5.6).

3.3.8 Bacteria and fungus resistance. The adhesive shall resist the growth of bacteria or fungus when tested as specified in 4.5.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

3.3.9 Condition in container. When tested as specified in 4.5.12 the adhesive shall be free from foreign matter and segregation or settling to such an extent as to render redispersion difficult.

3.3.10 Peel strength. When tested as specified in 4.5.10, the separation shall be in the adhesive and not in the paper.

3.4 Marking. Identification marking shall be in accordance with MIL-STD-130. The markings shall be applied to the containers, so that normal handling will not impair the legibility of the marking. Each container shall be marked with the date of manufacture. The expiration date of the adhesive shall be marked on each container.

3.5 Workmanship. The adhesive shall be a clean, uniform product free from foreign matter and processed in accordance with the best practice for high quality material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. When specified (see 6.2), test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article (see 4.3).
(b) Quality conformance (see 4.4).
(c) Packaging (see 4.6).

4.3 First article inspection. When specified (see 6.2), first article inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.3.1 First article sample. Statistical sampling and inspection of the first article shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective for major defects and 4.5 percent defective for minor defects.

4.3.2 First article inspection routine. The first article sample(s) shall be subjected to inspection and tests specified in Table I in the order shown.

TABLE I - Tests and Inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A		
Visual examination- - - - -	3.3	4.3.3
Transparency- - - - - - - -	3.3.6	4.5.11
Container Condition - - - -	3.3.9	4.5.12
Marking - - - - - - - - - -	3.4	5.3
Group B		
PH Value- - - - - - - - - -	3.3.2	4.5.2
Odor- - - - - - - - - - - -	3.3.3	4.5.3
Toxicity- - - - - - - - - -	3.3.4	4.5.4
Brushing Properties - - - -	3.3.5	4.5.5
Homogeneity - - - - - - - -	3.3.7	4.5.6
Fungus Resistance - - - - -	3.3.8	4.5.7
Storage Life- - - - - - - -	3.4	4.5.8
Chemical and Physical Requirements- - - - - - -	3.3	4.5.9
Peel Strength - - - - - - -	3.3.10	4.5.10

4.3.3 Failures. Allowable failures shall be as specified in Table II. One or more major defects shall be cause for refusal to grant first article approval.

TABLE II - Classification of Defects.

Examination	Defect(s)	Classification Major	Minor
Adhesive	Foreign Matter	X	
	Not homogeneous	X	
Container	Improper closure	X	
Marking	Omitted, incorrect or illegible	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of group A and B inspections specified in Table I. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for normal inspection, single sampling. The inspection level shall be II and the AQL shall be 1.5.

4.4.1 Inspection lot. An inspection lot shall consist of adhesive produced under essentially the same conditions, and offered for inspection at the same time.

4.4.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units as applicable, and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be reinspected, using tightened inspection. A lot shall not be resubmitted more than one time.

4.5 Tests. Examination and tests shall be performed as specified herein. Other test methods may be substituted upon written approval by the procuring activity, provided the supplier has demonstrated that the results obtained properly correlate with those obtained using the methods specified herein. The supplier must also demonstrate that such substitution does not relax the requirements specified herein.

4.5.1 Test conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-810.

4.5.2 pH value. The pH value shall be determined potentiometrically, using a pH meter with a glass electrode. The starch shall have a pH value between 3.6 and 7.0 as determined according to ASTM E7077.

4.5.3 Odor. The odor of the dried film of adhesive shall be determined by the use of the filter paper technique described in method 4401 of FED-STD-141 for compliance with 3.3.3.

4.5.4 Toxicological data and formulations. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.5.5 Brushing properties. The brushing and covering qualities of the mixed adhesive shall be determined by application to a 2 feet by 2 feet (0.609m by 0.609m) cold-rolled steel panel with a good quality 2-1/2-inch (63.5 mm) paint brush. The panels shall be at stabilized temperatures of 10°F (-12°C) increments from 50°F to 100°F (10°C to 38°C). The adhesive shall show good brush covering qualities and adhesive capabilities at all increments of specified temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

4.5.6 Consistency. For the consistency test, a sheet of white bond paper conforming to type II of UU-P-121 shall be covered with a thin layer of adhesive. The sheet shall be laid face down on a piece of glass and rolled with a paper or print roller sufficiently to eliminate air bubbles and wrinkles. Then the sheet shall be carefully gone over by hand for any unevenness caused by grit or undissolved matter to determine compliance with 3.3.7.

4.5.7 Fungus resistance. The procedures prescribed in ANSI G 2170 shall be used. The test specimens shall consist of dried adhesive 0.02 inch thick (0.51mm).

4.5.8 Storage. Samples of the adhesive shall be subjected to 10 days of storage at 68°F to 77°F (20°C to 25°C). The samples shall then be subjected to the quality conformance inspection and tests specified herein.

4.5.9 Chemical and physical requirement. The starch shall have a maximum limit of ash - 15 percent, protein 0.5 percent, and moisture 13 percent. Particle size shall not be less than 98 percent when passed through a U. S. standard number 100 screen.

4.5.10 Peel strength. The peel strength test shall conform to ANSI D903 using paper conforming to type I of MIL-P-10831. Evenly distribute a film of adhesive near the middle of a strip of paper, fold the strip upon itself, rub three times with the hand and allow the sample to dry for 4 hours. At the end of the drying period, pull apart the free ends of the strip with the same motion as in opening a book to determine compliance with 3.3.10.

4.5.11 Translucence. Samples shall be translucent with no discoloration. Test specimens shall be prepared as specified in 4.5.6.

4.5.12 Examination of filled containers. Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or less than required fill, shall not be offered for delivery. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, that will be cause for rejection of the lot represented by the sample.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements specified in Section 5. The sample plan

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

shall be in accordance with MIL-STD-105 for normal inspection, single sampling. The inspection level shall be S-4 and the AQL shall be 2.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level B or C as specified (see 6.2).

5.1.1 Level B. The adhesive shall be packaged in 1-pound (net weight) packages. The unit packages shall be sift proof sealed paperboard boxes conforming to PPP-B-566, style I, type A, or paper bags conforming to UU-B-36. Boxes shall have all seams completely sealed with tape conforming to PPP-T-76. Bags shall be sealed with adhesive or with tape conforming to PPP-T-76.

5.1.2 Level C. The adhesive shall be packaged in accordance with the manufacturer's commercial practice. Packages shall be sealed in a manner to prevent loss of contents.

5.1.3 Intermediate packaging. Adhesive, packaged as specified in 5.1.1., may be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape, and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level B. Fifty (50) 1-pound packages of adhesive shall be uniformly packed in fiberboard boxes conforming to PPP-B-636, class weather resistant. Boxes shall be sealed in accordance with the appendix thereto.

5.2.2 Level C. The adhesive, packaged as specified, shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This adhesive is intended for adhering paper targets to cloth targets.

6.2 Ordering data. Purchasers should excercise any desired options offered herein, and acquisition documents should specify the following:

6.2.1 Acquisition requirements

(a) Title, number and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

(b) When first article inspection is required (see 3.1)
(c) When test equipment and inspection facilities are required (see 4.1.1).
(d) Level of preservation, packaging and packing required (see 5.1 and 5.2)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data Requirements	Applicable DID no.	Options
4.3	First Article Inspection Procedure	DI-T-4901	-------
4.3	First Article Inspection Report	DI-T-4902	-------
4.4	Production/Acceptance Inspection Procedures	DI-T-4903	-------
4.4	Production Inspection Reports	DI-T-4904	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. Any changes or deviations from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish adhesive conforming to this specification.

6.4 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - MC
Army - MI

Review Activity:
Army - MI

Preparing Activity:
Navy - MC

Project No. 8040-0417

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-A-17682E ADHESIVE, STARCH

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-LF-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:27Z
Check the source to verify that this is the current version before use.